Biomedical Materials & Devices (2026) 4:388–413
https://doi.org/10.1007/s44174-025-00295-4

REVIEW

A Comprehensive Review on the Green Synthesis of Nanoparticles: Advancements in Biomedical and Environmental Applications

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Received: 29 October 2024 / Accepted: 10 February 2025 / Published online: 24 February 2025
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Abstract
Green synthesis of nanoparticles has gained significant attention as an eco-friendly and sustainable alternative to conventional chemical and physical methods, leveraging biological resources such as plants, bacteria, fungi and algae. This approach minimizes the use of hazardous chemicals, reduces energy consumption and improves the biocompatibility of nanoparticles, making it particularly suitable for biomedical and environmental applications. This review delves into recent advancements in green synthesis techniques, with a specific focus on the biomolecules involved in stabilizing and enhancing the properties of nanoparticles. The biomedical applications of green synthesized nanoparticles are of particular interest, with promising uses in areas such as drug delivery, bio imaging, antimicrobial therapies and cancer treatment. These nanoparticles offer advantages in terms of targeted delivery, reduced toxicity and improved therapeutic efficacy compared to traditional methods. The review also highlights the challenges faced in scaling up green synthesis processes, ensuring reproducibility, and comprehensively understanding the complex mechanisms behind the formation and functionalization of these nanoparticles. Furthermore, it proposes future research directions to overcome these challenges, including the optimization of synthesis protocols, deeper mechanistic studies and the development of more robust and reproducible techniques. In conclusion, the review underscores the transformative potential of green synthesized nanoparticles in addressing global health and environmental challenges.

Keywords Green synthesis · Nanoparticles · Eco-friendly · Biomedical applications · Biocompatibility · Sustainable nanotechnology

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Introduction

Nanotechnology: Potential and Challenges

Nanotechnology has emerged as a transformative force across a wide array of fields, from medicine and electronics to energy and environmental science [1]. At the heart of this technological revolution are nanoparticles, which are typically defined as particles ranging from 1 to 100 nm in size. Their unique physicochemical properties such as an exceptionally high surface area, increased reactivity and enhanced mechanical strength set them apart from bulk materials and contribute to their diverse range of applications [2]. These characteristics enable nanoparticles to provide innovative solutions to some of the most pressing global challenges, including disease treatment, pollution management and energy efficiency. As such, nanoparticles are poised to play a crucial role in shaping the future of technology [3]. The

history of modern nanotechnology traces back to Michael Faraday's pioneering work on colloidal gold particles in the nineteenth century, which laid the foundation for later advancements in nanoparticle research [4]. This trajectory was further accelerated by Richard Feynman's visionary 1959 lecture, where he outlined the potential of manipulating matter at the atomic scale, and expanded by Eric Drexler, who coined the term "nanotechnology" in the 1980s to describe the manipulation of matter at the molecular level [5]. The practical application of nanoparticles gained significant momentum in the 1990s, following breakthroughs in imaging technologies such as atomic force microscopy and scanning tunneling microscopy, which allowed for the visualization and manipulation of individual nanoparticles [6].

One of the most promising and environmentally conscious approaches to nanoparticle synthesis is the green synthesis method, which utilizes biological agents such as plant extracts, fungi, and bacteria [7]. This eco-friendly method offers a sustainable alternative to traditional chemical and physical synthesis methods by reducing the use of toxic chemicals and improving the biocompatibility of the nanoparticles. However, challenges remain in optimizing the green synthesis process, particularly concerning factors such as precursor concentration, pH, reaction time, temperature and the variability of biological reducing agents [8]. These factors can significantly affect the consistency, size, shape and functionality of the nanoparticles, posing a barrier to large-scale, reproducible production. Despite growing interest in green synthesis, much of the research thus far has focused primarily on a limited range of nanoparticles, such as silver, copper, zinc and iron, with less attention given to the broader factors that influence nanoparticle formation [9]. There is a need for more comprehensive research that addresses the complex interplay of biological, chemical and environmental variables in the green synthesis process. Moving forward, the focus of nanoparticle development will need to emphasize sustainable practices that minimize environmental impact while expanding the scope of their applications [10]. Green synthesis holds enormous potential in fields like targeted drug delivery, renewable energy solutions, advanced material science and water purification, all of which could greatly benefit from nanotechnology innovations [6].

Addressing the challenges of optimizing and standardizing green synthesis techniques will require interdisciplinary collaboration across fields such as chemistry, biology and engineering [11]. Developing standardized protocols and frameworks will be critical for the widespread application of green-synthesized nanoparticles [12]. This review explores the evolving landscape of nanoparticle research, examining the opportunities and challenges in sustainable synthesis and their application across various industries [13]. While research interest in green synthesis has grown considerably over the past decade, many studies still focus narrowly on specific nanoparticles, overlooking the broader environmental and practical factors at play [14]. The findings and novelty of this study aim to make a significant contribution to the field by investigating a wide array of biological agents and varying synthesis conditions. By doing so, the study seeks to identify optimal strategies for advancing sustainable nanotechnology [15]. Through a comprehensive exploration of different biological resources and synthesis parameters, this research intends to enhance the efficiency, scalability and environmental compatibility of nanoparticle production, thereby fostering more eco-friendly and innovative applications of nanotechnology in various fields.

Nanoparticle Classifications

Nanoparticles can be classified based on various criteria, each offering a unique perspective on their properties and applications. These classifications include dimensionality, chemical composition, morphology and synthesis methods, as depicted in Fig. 1; Table 1. The chemical composition of nanoparticles plays a critical role in determining their purity, stability and overall performance, which is essential for their applications in various fields such as medicine, electronics and environmental science [16, 17].

Classification by Composition

The composition of nanoparticles is one of the most significant factors influencing their functionality, performance, and specific applications. The most common categories based on composition include metallic, metal oxide, carbon based, polymeric, ceramic and lipid-based nanoparticles [18].

Metal-Based Nanoparticles

Metallic nanoparticles, such as those composed of gold, silver, platinum and copper, are among the most extensively studied and widely used nanoparticles. These materials are highly valuable in a variety of applications due to their excellent electrical conductivity, catalytic activity and biocompatibility [19]. In the metal-based nanoparticles offer large surface areas and high reactivity, making them ideal for accelerating chemical reactions in industrial processes. In medicine, these nanoparticles have shown promise in applications such as drug delivery, cancer therapy and diagnostic imaging due to their ability to be easily functionalized with targeting molecules [20].

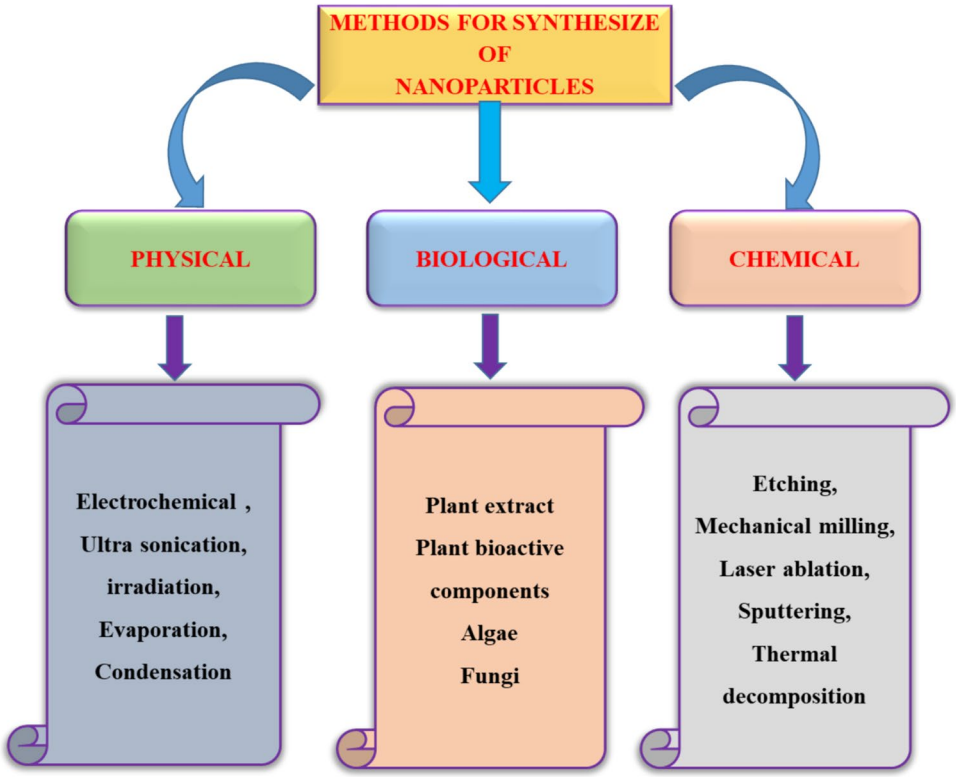

Fig. 1 Green synthesized nanoparticles different methods

Table 1 Properties of carbon-based, metal-based and metal oxide-based nanoparticles

Nanoparticle type	Nanoparticles	Properties	References
Carbon-based nanoparticles	Carbon nanotubes	High electrical and thermal conductivity	[4–12]
	Carbon nanofiber	High thermal and electrical properties, frequency shielding, mechanical strength	
	Fullerenes	Safe and inert, semiconductor, conductor, superconductor, light intensity transmission	
	Carbon black	High strength, electrical conductivity, UV degradation resistance	
	Graphene	Extreme strength, light absorption	
Metal-based nanoparticles	Silver	Light absorption and scattering, stable, antibacterial, disinfectant	
	Zinc	Antibacterial, anti-corrosive, antifungal, UV filtering	
	Iron	Reactive, unstable, sensitive to air, water	
	Copper	Ductile, very high thermal and electrical conductivity, highly flammable solids	
	Gold	Reactive, interacts with visible light	
	Cobalt	Magnetic, toxic, unstable, absorbs microwaves	
	Cadmium	Semiconductor of electricity, insoluble	
	Lead	Highly toxic, reactive, stable	
	Aluminum	High reactivity, heat and sunlight, large surface area	
Metal oxide-based nanoparticles	Zinc oxide	Antibacterial, anti-corrosive, antifungal, UV filtering	
	Titanium oxide	High surface area, magnetic, inhibits bacterial growth	
	Iron oxide	Reactive, unstable	
	Cerium oxide	Antioxidant, low reduction potential	
	Aluminum oxide	High reactivity, sensitive to moisture	
	Silicon dioxide	Stable, less toxic	
	Magnetite	Magnetic, highly reactive	

Metal Oxide-Based Nanoparticles

Metal oxide nanoparticles, including titanium dioxide (TiO_2), zinc oxide (ZnO) and iron oxide (Fe_3O_4), have diverse applications in fields ranging from environmental protection to healthcare [21]. Titanium dioxide nanoparticles, for instance, are widely used as photocatalysts in the degradation of environmental pollutants, while zinc oxide is commonly used in sunscreens and cosmetics due to its UV blocking properties [22]. Iron oxide nanoparticles, including magnetite (Fe_3O_4), have gained considerable attention for their role in targeted drug delivery and magnetic resonance imaging (MRI), as well as in environmental cleanup, particularly in the removal of heavy metals from water. Metal oxide nanoparticles are favored for their stability, non-toxicity and multifunctionality [23].

Carbon Based Nanoparticles

Carbon based nanoparticles, such as fullerenes, carbon nanotubes (CNTs), grapheme and carbon dots, have garnered significant interest due to their unique properties, including exceptional electrical conductivity, mechanical strength and chemical stability [24]. Graphene, a single layer of carbon atoms arranged in a two dimensional lattice, is particularly notable for its outstanding thermal and electrical conductivity, making it a key material for electronic devices, sensors and energy storage systems [25]. Carbon nanotubes, which are cylindrical structures of carbon atoms, exhibit remarkable tensile strength and are used in applications ranging from electronics to drug delivery. Fullerenes (C_{60}) are used in drug delivery and imaging due to their unique ability to encapsulate other molecules. Carbon dots, another emerging class of carbon based nanoparticles, are being explored for use in fluorescent bio imaging and as carriers for drug delivery systems [26].

Polymeric Nanoparticles

Polymeric nanoparticles are composed of synthetic or natural polymers and are widely employed in biomedical applications, particularly in drug delivery systems. The use of polymers in nanoparticle synthesis offers several advantages, including ease of functionalization, biocompatibility and controlled release properties [27]. Polymeric nanoparticles can be engineered to encapsulate a variety of therapeutic agents, such as chemotherapeutic drugs or proteins, allowing for targeted and sustained release [28]. These nanoparticles also offer the ability to improve the bioavailability of poorly soluble drugs and reduce side effects by providing targeted delivery to specific tissues or organs. The commonly used polymers include poly lactic-co-glycolic acid (PLGA) and chitosan, which are known for their biodegradability and safety in clinical applications [29].

Ceramic Nanoparticles

Ceramic nanoparticles, including silica (SiO_2) and alumina (Al_2O_3), are non-metallic inorganic materials that offer unique properties such as high surface area, chemical stability, and mechanical strength [30]. Silica nanoparticles, in particular, have widespread use in catalysis, as they can provide large surface areas for reactions. They are also used in coatings, drug delivery, and as fillers in composite materials [31]. Alumina nanoparticles, known for their hardness and wear resistance, are employed in the production of advanced materials and coatings that require high durability. Ceramic nanoparticles are also being investigated for use in environmental remediation and water purification due to their high porosity and surface reactivity [32].

Lipid-Based Nanoparticles

Lipid-based nanoparticles, such as liposomes, solid lipid nanoparticles (SLNs), and nanostructured lipid carriers (NLCs), are primarily used in pharmaceutical applications for drug delivery [33]. These nanoparticles are composed of lipids, which are naturally occurring molecules that are biocompatible and able to encapsulate both hydrophobic and hydrophilic compounds. Lipid nanoparticles are particularly useful for delivering poorly water soluble drugs and ensuring their controlled release within the body [34]. Liposomes, in particular, are spherical vesicles that can encapsulate drugs, proteins and genetic material and are used in various drug delivery systems, including vaccines and gene therapy. Solid lipid nanoparticles and nanostructured lipid carriers offer advantages in terms of stability and drug loading capacity and are being investigated for use in both systemic and targeted drug delivery [35].

Classification by Morphology

The morphology of nanoparticles plays a critical role in determining their unique physical, chemical and biological properties. The shape, size and structure of nanoparticles can influence their interaction with surrounding environments, including biological systems, materials and other nanoparticles [36]. Morphological properties of nanoparticles are influenced by several factors, including the type of material used, the specific synthesis methods such as green synthesis and the presence of various biological agents or structures in the synthesis medium, such as leaf extracts [37]. The morphology of nanoparticles is also influenced by the synthesis conditions, such as temperature, pH, precursor

concentration and time, as well as the use of surfactants or stabilizing agents to control particle formation [38].

Dimensionality of Nanoparticles

Nanoparticles can be classified based on their dimensionality, which refers to the number of spatial dimensions in which their properties are constrained. The dimensionality of nanoparticles significantly influences their behavior, including how they interact with light, heat and other substances [39].

0D Nanoparticles These nanoparticles are dimensionally confined in all directions. Quantum dots and spherical nanoparticles are examples of 0D nanoparticles. Due to their small size and the quantum effects at the nanoscale, 0D nanoparticles exhibit unique optical and electronic properties, such as fluorescence and size dependent energy band gaps. Quantum dots, for instance, are used extensively in imaging, sensors and optoelectronics [40].

1D Nanoparticles These are nanoparticles where one dimension is significantly larger than the others, leading to elongated structures. Nano rods and nanowires are typical examples of 1D nanoparticle. These materials exhibit anisotropic properties, meaning that their properties vary depending on the direction of measurement. Nano rods are known for their unique optical properties and are used in applications like surface plasmon resonance sensing, drug delivery and catalysis [41].

2D Nanoparticles These nanoparticles have two dimensions that are much larger than the third dimension. Examples include nanosheets and graphene, where the material is confined to a thin layer in one dimension while maintaining large lateral dimensions in the other two. 2D materials, such as graphene, are well known for their exceptional electrical, mechanical and thermal properties, making them ideal for applications in electronics, energy storage and composite materials [42].

3D Nanoparticles 3D nanoparticles have significant dimensions in all directions and often feature complex geometries. Dendrimers, nanocubes and nanoporous materials are examples of 3D nanoparticles. These structures can have high surface areas and multiple functional sites, allowing for applications in drug delivery, sensors and catalysis, where surface interactions are crucial [43].

Size of Nanoparticles

The size of nanoparticles is a crucial factor in characterizing their properties, as it directly impacts their physical, chemical and biological behaviours. Nanoparticles, typically ranging from 1 to 100 nm (nm), exhibit unique characteristics that bulk materials do not, such as enhanced reactivity, increased surface area and the ability to interact more efficiently with biological systems [44]. The small size allows nanoparticles to penetrate biological barriers, interact with cellular structures and exhibit behaviors like quantum effects that are not seen in larger particles. Consequently, accurate measurement of nanoparticle size is essential for tailoring their properties for specific applications in fields such as drug delivery, diagnostics and materials science [45].

Scanning Electron Microscopy (SEM) and Transmission Electron Microscopy (TEM) Both SEM and TEM are powerful imaging techniques that provide highly detailed information about nanoparticle size, morphology and structure. These methods are particularly valuable because they allow direct visualization of nanoparticles at the nanometer scale, offering insights into their shape, surface features and internal architecture [46]. SEM provides high-resolution images of the surface morphology of nanoparticles. It works by scanning the surface of a sample with a focused beam of electrons and capturing the secondary electrons emitted from the surface. SEM is excellent for observing surface structures and can reveal important details about surface roughness, porosity and particle shape [47]. Although it provides detailed surface information, SEM typically requires the sample to be coated with a conductive material, which may alter the nanoparticle's properties. TEM, on the other hand, offers even higher resolution, allowing for visualization of internal nanoparticle structures. This technique passes a beam of electrons through a very thin sample and the transmitted electrons are used to generate highly magnified images of both the surface and internal structures [48]. TEM can provide information on the particle's internal composition, crystallinity and even atomic arrangement. This makes TEM particularly useful for analyzing nanoparticles with complex internal structures or coatings. However, TEM requires the sample to be very thin and can be more time consuming and technically challenging compared to SEM analysis [49].

Laser Diffraction Laser diffraction is a widely used technique for measuring the size distribution of solid nanoparticles. In this method, a laser beam is passed through a sample and the scattering pattern of the light is analyzed to determine the particle size distribution [50]. The angle and intensity of the scattered light are directly related to the size of the particles, smaller particles scatter light at larger angles, while larger particles scatter at smaller angles. This method is particularly effective for bulk measurements, offering rapid and efficient analysis of nanoparticle population. It is often used in industries where quick assessment of particle

size is necessary, such as in pharmaceuticals or materials science [51].

Photon Correlation Spectroscopy (PCS) Also known as Dynamic Light Scattering (DLS), Photon Correlation Spectroscopy is a technique used to measure the size of nanoparticles in a liquid environment. It analyzes the fluctuations in the intensity of scattered light as it interacts with nanoparticles in suspension. These fluctuations occur because of the Brownian motion of particles and the rate of these fluctuations is inversely proportional to the size of the particles [52]. PCS provides valuable information about the hydrodynamic diameter of nanoparticles, which is an important parameter that accounts for the particle's size in solution, including the effects of surface coatings and possible aggregation. This technique is particularly useful for characterizing nanoparticles in colloidal suspensions and is widely employed in research on drug delivery systems and nanomedicine [53].

Centrifugation Centrifugation is a versatile method used to separate nanoparticles based on differences in their size, density and shape by applying a centrifugal force. When a suspension containing nanoparticles is spun at high speeds, particles of different sizes experience different centrifugal forces, causing them to separate into distinct layers or fractions [54]. This technique is commonly used for isolating nanoparticles from complex mixtures, purifying specific particle sizes, or determining the size distribution within a suspension. In addition to its ability to separate nanoparticles, centrifugation can also be applied to concentrate nanoparticles, remove impurities and assess the stability of nanoparticle suspensions over time. This method is particularly beneficial for the analysis of larger particle sizes or heterogeneous nanoparticle populations [54].

Morphological Types of Nanoparticles

Nanoparticles can take various shapes, each of which imparts specific properties that are useful for different applications. Some of the most common nanoparticle shapes include,

Spherical Nanoparticles Spherical nanoparticles are the most common and simplest form of nanoparticles, having a roughly round shape. Gold and silver nanoparticles are classic examples of spherical nanoparticles. The uniformity of their shape allows for easy functionalization and consistent behavior across different applications. Spherical nanoparticles are widely used in drug delivery systems, diagnostics, and imaging and as catalysts. The large surface area to volume ratio of spherical nanoparticles enhances their reactivity and provides a high surface area for drug attachment [55].

Rod-Shaped Nanoparticles (Nano rods) Nanorods are cylindrical nanoparticles, where one dimension (length) is significantly longer than the other two (diameter). Gold nanorods are one of the most studied examples of this morphology. Nanorods exhibit anisotropic properties, meaning they have different behaviors along different axes. This anisotropy can influence optical properties such as plasmonic resonance, making them useful for applications in sensing, imaging and photothermal therapy. Nanorods can also be used in drug delivery, where their elongated shape may improve cellular uptake compared to spherical nanoparticles [56].

Tubular Nanoparticles (Nanotubes) Nanotubes are hollow, cylindrical structures that can be made from a variety of materials, including carbon, silica and metal oxides. Carbon nanotubes (CNTs) are the most famous example and are known for their remarkable mechanical, electrical and thermal properties. CNTs are extensively used in electronics, nanocomposites and energy storage applications. The hollow structure of nanotubes also makes them an ideal platform for loading and delivering drugs or other bioactive molecules. The mechanical strength and flexibility of nanotubes are also advantageous for various structural and engineering applications [57].

Platelet Shaped Nanoparticles Platelet-shaped nanoparticles are flat and two dimensional, resembling thin plates. Graphene and nanoclays are common examples of this morphology. These nanoparticles are useful for applications where high surface area, mechanical strength and conductivity are required. Graphene, for widely used in energy storage devices, such as super capacitors and batteries, as well as in composite materials to enhance their strength and conductivity. Platelet nanoparticles are also used in drug delivery, where their large surface area allows for efficient loading of therapeutic agents [58].

Core–Shell Nanoparticles Core–shell nanoparticles consist of a core material encapsulated by a shell of a different material. This morphology allows for the combination of the properties of both materials. For instance, a core of magnetic nanoparticles such as iron oxide could be coated with a biocompatible shell, allowing for targeted drug delivery and magnetic resonance imaging (MRI) applications. Core shell nanoparticles are also widely used in catalysis, where the core material provides catalytic activity, while the shell material can offer stability and selectivity. In drug delivery, core shell nanoparticles can be engineered for controlled release and protection of sensitive drugs [59].

Nanoparticle Green Synthesis Methods

The green synthesis of nanoparticles has emerged as a sustainable and eco-friendly alternative to conventional synthetic methods. This approach leverages natural agents such as plant extracts, microorganisms and biomolecules to produce nanoparticles, offering advantages such as reduced toxicity, improved biocompatibility and minimized environmental impact [60]. Green synthesis methods can be broadly categorized into two main approaches such as top down and bottom-up methods, as illustrated in Fig. 2**;** Table 2. These methods differ in how nanoparticles are formed and in the techniques used to control their size, shape and properties.

Top-Down Methods

Top down methods involve breaking down bulk materials into nanoscale particles through mechanical or physical processes. These methods are often used in traditional manufacturing industries and have been adapted for nanoparticle production. Some of the key top-down methods such as,

Mechanical Milling Mechanical milling involves the use of mechanical energy to grind bulk materials into fine nanoparticles. Ball milling is a common technique that uses steel or ceramic balls to break down larger particles. This method is particularly useful for producing metal and alloy nanoparticles, especially when large quantities are required. However, while the process is scalable, it often results in uneven particle size distribution and may introduce defects, such as surface roughness or contamination from milling media [61].

Lithography Lithography methods, including electron beam lithography, photolithography and Nano imprint lithography, are employed to create nanoscale patterns on substrates, primarily for semiconductor fabrication and

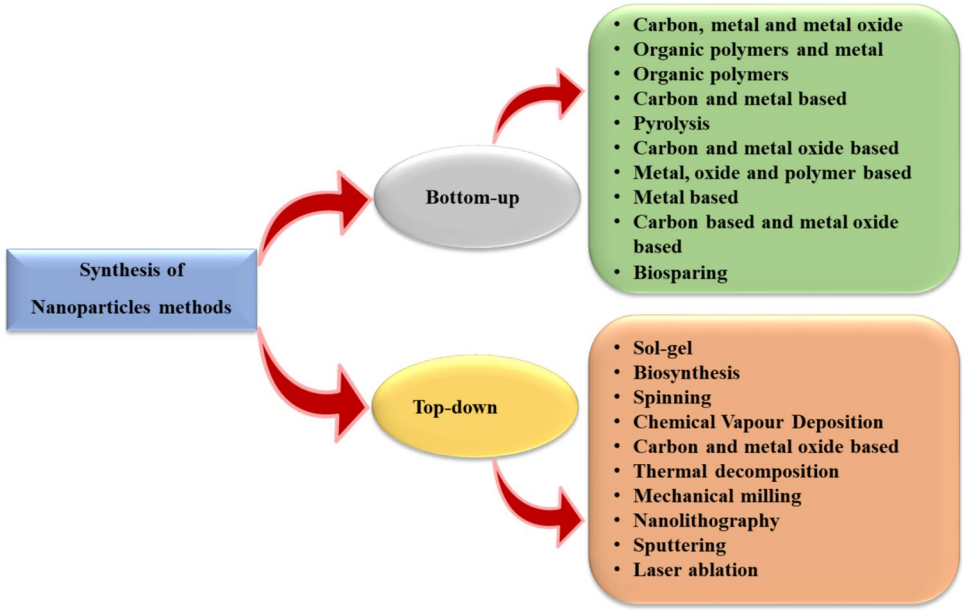

Fig. 2 Different type of methods in nanoparticles

Table 2 Different types of synthesized nanoparticles based on the approach in Bottom-up and Top-down methods

Approach	Nanoparticle types	Methods	References
Bottom-up	Carbon, metal, and metal oxide	Sol–gel	[15–23]
	Organic polymers and metal	Biosynthesis	
	Organic polymers	Spinning	
	Carbon and metal-based	Chemical vapor deposition	
	Carbon and metal oxide-based	Pyrolysis	
Top-down	Carbon and metal oxide-based	Thermal decomposition	
	Metal, oxide, and polymer-based	Mechanical milling	
	Metal-based	Nanolithography	
	Carbon and metal oxide-based	Sputtering	
	Metal-based	Laser ablation	

Nano electronics. These methods allow for high precision and are often used to create intricate nanostructures with controlled dimensions. However, lithography is expensive, complex and generally limited by the materials that can be processed using these techniques, making them less ideal for large scale nanoparticle production [62].

Laser Ablation Laser ablation involves the use of high intensity laser pulses to remove material from a solid target, producing nanoparticles in the process. This method can create high purity metal, ceramic, and composite nanoparticles, as the laser allows for the direct ablation of bulk materials into the vapor phase, followed by rapid cooling and nanoparticle formation. Despite the high quality nanoparticles produced, laser ablation requires sophisticated equipment, making it impractical for large scale production [63].

Etching Etching techniques, both chemical and physical, are used to remove material from a substrate to create nanostructures. Wet etching typically involves chemical reactions with a solvent, while dry etching uses ionized gases or plasmas to etch materials. Etching is widely used in the fabrication of microelectronics, photonics and nanostructured surfaces. While it offers high precision, it is a time consuming process that requires expensive equipment and may be less suitable for large scale production of nanoparticles [64].

Bottom-Up Methods

Bottom-up methods, in contrast, involve the assembly of nanoparticles from molecular or atomic precursors. These methods allow for precise control over the size, shape and composition of nanoparticles, enabling the production of highly uniform and tailored particles [65]. Some of the primary bottom-up green synthesis methods include,

Chemical Vapor Deposition (CVD) Chemical Vapor Deposition (CVD) is a technique in which gaseous precursors decompose on a heated substrate to form nanoparticles. This method is commonly used to produce carbon nanotubes, graphene and other nanostructures. CVD allows for high purity, uniformity and scalability, making it an attractive option for large-scale production. However, it requires high temperatures often exceeding 800 °C and tightly controlled conditions to ensure successful deposition and nanoparticle formation. The process can also be energy intensive, limiting its eco-friendliness when compared to other green synthesis methods [66].

Sol–Gel Synthesis The sol–gel process involves the transformation of precursor materials into a gel-like phase, which is then dried and calcined to form nanoparticles. Sol–gel synthesis is widely used for the production of ceramic and metal oxide nanoparticles, such as silica (SiO_2), titanium dioxide (TiO_2), and zinc oxide (ZnO) [67]. This method is highly versatile, allowing for the production of nanoparticles with precise control over composition and size. However, sol–gel synthesis requires careful control of reaction parameters, including temperature, pH and precursor concentration, to achieve optimal results [68].

Chemical Reduction Chemical reduction is a simple and cost-effective method for synthesizing metallic nanoparticles, including gold, silver and copper nanoparticles. In this process, metal salts are reduced by reducing agents such as sodium citrate, hydrogen and plant-based agents [69]. This method is highly favored for its simplicity and versatility, as it does not require high temperatures and complex equipment. However, it is important to control the reducing agent concentration and other reaction parameters to ensure uniform size and high quality nanoparticles [70].

Self-Assembly Self-assembly is a process where molecules spontaneously organize themselves into nanoscale structures without the need for external direction. This process can be guided by intermolecular interactions such as van der Waals forces, hydrogen bonding or electrostatic interactions [71]. In nanoparticle synthesis, self-assembly is used to organize molecules into quantum dots, micelles or other nanoscale structures. While self-assembly offers a high degree of precision, it is material-specific, often slow and difficult to scale for large scale production. The success of this method depends heavily on the choice of materials and reaction conditions [72].

Hydrothermal and Solvothermal Synthesis Hydrothermal and solvothermal synthesis methods involve the use of high temperature and high-pressure conditions to produce nanoparticles. In hydrothermal synthesis, water is the solvent, whereas in solvothermal synthesis, organic solvents are used. These methods are highly effective for synthesizing metal oxide, carbon based and composite nanoparticles [73]. They offer excellent control over particle size and purity but require specialized equipment capable of withstanding the extreme conditions involved. Additionally, while these methods can produce high-quality nanoparticles, they are often energy-intensive and may not be as environmentally friendly as other green synthesis techniques [74].

Biological Synthesis and Hybrid Approaches

Biological synthesis of nanoparticles has gained significant attention as a sustainable and eco-friendly alternative to traditional chemical and physical methods. This approach leverages natural agents such as microorganisms such as bacteria, fungi, plant extracts and enzymes to produce

nanoparticles [75]. These biological systems play a crucial role in nanoparticle synthesis by reducing metal ions into nanoparticles, often leading to products with unique properties that differ from those synthesized by conventional methods [76]. Notable examples of biologically synthesized nanoparticles include gold, silver, copper and iron oxide nanoparticles. These nanoparticles offer several advantages, such as enhanced biocompatibility, reduced toxicity and a lower environmental impact, making them highly suitable for applications in fields like medicine, electronics, and environmental remediation [77]. However, biological synthesis methods, although environmentally friendly, face challenges, particularly in terms of scalability and speed. The synthesis process typically occurs over a longer period of time, and factors such as temperature, pH, the concentration of biological agents, and the presence of other compounds can influence the size, shape and yield of the nanoparticles produced [78]. Additionally, controlling the uniformity of the nanoparticles produced by biological processes remains a significant challenge, as the complexity of the biological systems involved can lead to variations in the particle characteristics [79].

Despite these challenges, the development of hybrid approaches that combine the strengths of both top down and bottom up techniques offers promising solutions. Hybrid approaches aim to address the limitations of biological synthesis by integrating the precision and scalability of physical or chemical methods with the sustainability and biocompatibility of biological processes [80]. For the, combining biological agents like plant extracts with mechanical milling or laser ablation techniques allows for the fine tuning of nanoparticle properties, such as size, shape and surface functionalization. This hybridization enhances the control over nanoparticle synthesis, making it possible to achieve more precise nanostructures suitable for specific applications, such as drug delivery or electronic devices [81]. However, the complexity and cost of these hybrid methods remains a barrier to widespread adoption. In the realm of Nano medicine, the integration of medicinal plants with nanoparticles holds immense potential for advancing drug delivery, therapeutics and diagnostic applications [82]. Many medicinal plants are rich in bioactive compounds, such as alkaloids, flavonoids and phenolic acids that not only act as reducing agents for the synthesis of nanoparticles but also impart therapeutic properties to the nanoparticles themselves [83]. By utilizing plant extracts in nanoparticle synthesis, researchers can create hybrid nanomaterials that offer enhanced bioavailability, targeted delivery and reduced side effects compared to traditional drugs. For instance, gold and silver nanoparticles synthesized using plant extracts have been shown to possess anti-inflammatory, antioxidant and antimicrobial properties, which can be leveraged for the treatment of various diseases [84].

Moreover, plant derived nanoparticles can improve the stability and solubility of drugs, which is often a challenge in conventional drug delivery systems. The bio functionalization of nanoparticles through plant compounds also enhances their interactions with biological systems, improving the efficacy of drug delivery and ensuring that the drug reaches the targeted site with minimal systemic toxicity [85]. The nanoparticles made from green synthesis methods have demonstrated enhanced cancer cell targeting, reducing the adverse effects typically associated with conventional chemotherapy. This green and sustainable approach to nanomedicine not only reduces the reliance on toxic chemicals but also makes the process of nanoparticle synthesis more biocompatible, providing a safer alternative for drug delivery [86]. Additionally, the use of plants, which are renewable resources, aligns with sustainable development goals and supports the growing demand for eco-friendly and cost effective alternatives in pharmaceutical production. In conclusion, the biological synthesis of nanoparticles represents a promising path toward environmentally friendly and sustainable nanomaterial [87]. While the slower reaction times and challenges in scalability remain key obstacles, hybrid approaches that combine the advantages of biological processes with conventional top down and bottom-up techniques offer potential solutions [88]. In particular, the integration of medicinal plants with nanoparticles is opening new avenues for nanomedicine, offering a greener, more effective approach to drug delivery and therapy. Moving forward, advancements in hybrid synthesis methods and a better understanding of the interactions between biological agents and nanoparticles will be key to overcoming current limitations and unlocking the full potential of these innovative materials [89].

Challenges and the Need for Green Synthesis

The synthesis of nanoparticles (NPs) has gained substantial attention due to the unique properties these materials offer, such as high surface area, reactivity, and mechanical strength. Traditional methods for synthesizing nanoparticles typically involve chemical or physical processes, which often come with significant drawbacks, including high costs, environmental pollution and the use of toxic materials [90]. These challenges are particularly evident in the case of metallic nanoparticles, where processes like high temperature reduction and chemical treatments require hazardous reagents and result in waste production, making them less sustainable. This highlights the growing need for ecofriendly and biocompatible alternatives, which has led to an increasing interest in green synthesis methods [91].

Biological Synthesis: Intracellular and Extracellular Approaches

Green synthesis of nanoparticles utilizes biological systems such as plants, fungi, bacteria, algae and viruses to synthesize nanoparticles in an environmentally friendly manner. These methods rely on the natural reductive properties of biological agents to reduce metal ions and form nanoparticles, offering several advantages over traditional methods, including sustainability, lower toxicity and enhanced biocompatibility [92]. Green synthesis is broadly categorized into intracellular and extracellular approaches such as Intracellular synthesis involves synthesizing nanoparticles within the living cells of microorganisms or plants. For instance, certain plants are capable of hyper accumulating metals, which are subsequently reduced to nanoparticle forms within the cytosol due to the plant natural reductive capacity [93]. The intracellular process allows for the synthesis of nanoparticles in a controlled environment, as the biological system regulates particle growth, resulting in nanoparticles that are often uniform in size and shape. On the other hand, extracellular synthesis occurs outside the living cells, typically in the surrounding medium. In this method, biological extracts, such as plant leaves or fungi, are mixed with metal salt solutions [94]. Biomolecules such as polyphenols, terpenoids and flavonoids found in these extracts act as reducing agents, donating electrons to metal ions and reducing them to their zero-valent state. This reduction forms crystalline nuclei that grow into nanoparticles. The biomolecules present in the extract also serve as capping agents, which stabilize the nanoparticles and prevent agglomeration, ensuring that the particles remain dispersible and functional [95]. The use of plant extracts in particular provides several advantages, as they are abundant, easily available and contain a wide variety of reducing agents and stabilizing compounds that aid in the nanoparticle synthesis process [96].

Challenges in Green Synthesis

While green synthesis offers promising benefits, it also faces several challenges that need to be addressed to fully realize it's potential. One major challenge is the variability in nanoparticle synthesis [97]. Factors such as the type of biological agent used the concentration of metal salts, pH, temperature and even the growth conditions of the organisms or plants can all significantly impact the size, shape and properties of the resulting nanoparticles. This variability can make it difficult to achieve the desired consistency and reproducibility, which are essential for large-scale applications [98]. Another limitation lies in the complexity of nanoparticle formation**,** especially when it comes to synthesizing metallic oxide nanoparticles. While bio-reduction is effective in reducing metal ions like $Zn2^{+}$, $Cu2^{+}$ and Ag^{+} to form metallic nanoparticles, the mechanisms behind the synthesis of metal oxides (e.g., ZnO, CuO) are not as well understood. For instance, the formation of metal oxides requires precise control over the oxidation states of metal ions and their interaction with the surrounding biological agents [99].

This complexity can make it difficult to control the morphology and stability of oxide based nanoparticles, which are critical for certain applications, such as catalysis and environmental remediation [100]. Scalability remains another challenge for green synthesis methods. While laboratory scale synthesis using biological agents can yield nanoparticles with desirable properties, scaling up these processes for industrial or commercial production remains difficult [101]. Factors such as the cost of biological agents, the time required for synthesis and the control of reaction conditions all complicate the transition from laboratory to large scale production. Additionally, scaling up the process often introduces inconsistencies in nanoparticle characteristics, such as size and shape, which can compromise their performance in real world applications [102].

The Growing Need for Green Synthesis

The traditional methods for nanoparticle synthesis often involve toxic chemicals, high energy consumption, and significant environmental impacts. In contrast, green synthesis offers a more sustainable and ecofriendly alternative by utilizing biological agents that can produce nanoparticles without the need for hazardous chemicals [103]. Biological agents such as bacteria**,** fungi**,** plants**,** algae and viruses are particularly advantageous due to their ability to synthesize nanoparticles under mild conditions, reducing the need for high energy input or toxic reagents [104]. These methods are not only more environmentally friendly but also lead to the production of nanoparticles that are biocompatible, making them suitable for biomedical applications. One of the key advantages of plant-based green synthesis is that plant extracts are rich in natural reducing agents, such as polyphenols and flavonoids, which are capable of reducing metal ions to their nanoparticle forms [105]. Plants are also readily available and the synthesis process is typically faster and more cost effective compared to using microorganisms. Furthermore, plant-based synthesis allows for the production of nanoparticles that are often more stable and have better-controlled size and morphology [106].

Despite these advantages, challenges in optimization persist, particularly in terms of reaction parameters, biological agent variability and scalability. To overcome these challenges, ongoing research is focused on improving process optimization, developing standardized protocols and exploring the potential for hybrid approaches that combine the benefits of both biological and traditional synthesis methods [107]. The increasing demand for sustainable

practices in nanotechnology underscores the need for green synthesis methods. By utilizing biological systems for nanoparticle production, we can mitigate the ecological and health risks posed by conventional chemical and physical synthesis methods [106]. While there are challenges in terms of scalability, consistency and complexity, the potential benefits of green synthesis in terms of reduced toxicity, enhanced biocompatibility and environmental sustainability make it a critical area of research for the future of nanotechnology. Efforts to optimize these methods and overcome the existing challenges will be key in advancing green nanotechnology toward widespread industrial and biomedical applications [108].

Types of Nanoparticles and Their Areas of Applications

Nanoparticles (NPs) are diverse in both composition and form, and their properties have led to a wide range of applications across multiple industries. These nanoparticles, both organic and inorganic, are instrumental in fields such as medicine, agriculture and environmental remediation and consumer products [109]. This section explores the different types of nanoparticles and their varied applications, which are illustrated in Fig. 3; Table 3.

Diagnostic Applications of Nanoparticles

Nanoparticles significantly enhance diagnostic capabilities due to their unique optical, electrical and chemical properties. They serve as powerful contrast agents in magnetic resonance imaging (MRI), computed tomography (CT) and optical imaging, making it possible to visualize internal organs and tissues with high precision [110]. For the gold nanoparticles (Au-NPs) are used to improve the contrast in CT scans, while super paramagnetic iron oxide nanoparticles (SPIONs) enhance the sensitivity of MRI scans, making it easier to detect diseases at an early stage [111]. In addition to imaging, nanoparticles are essential in the development of sensitive biosensors that detect specific biomarkers for diseases like cancer, cardiovascular diseases and infections. These sensors are crucial for early diagnosis, which can improve patient outcomes by allowing for timely intervention. Theranostic nanoparticles, which combine both diagnostic and therapeutic functions, are used to simultaneously diagnose diseases and deliver targeted therapies. These nanoparticles can be used in personalized medicine, where imaging techniques help monitor the effectiveness of treatment in real time [112]. Nanoparticles have revolutionized drug delivery systems by enhancing the bioavailability of drugs, reducing side effects, and allowing for targeted drug delivery. By encapsulating drugs in nanoparticle carriers, it becomes possible to deliver therapeutic agents directly to the

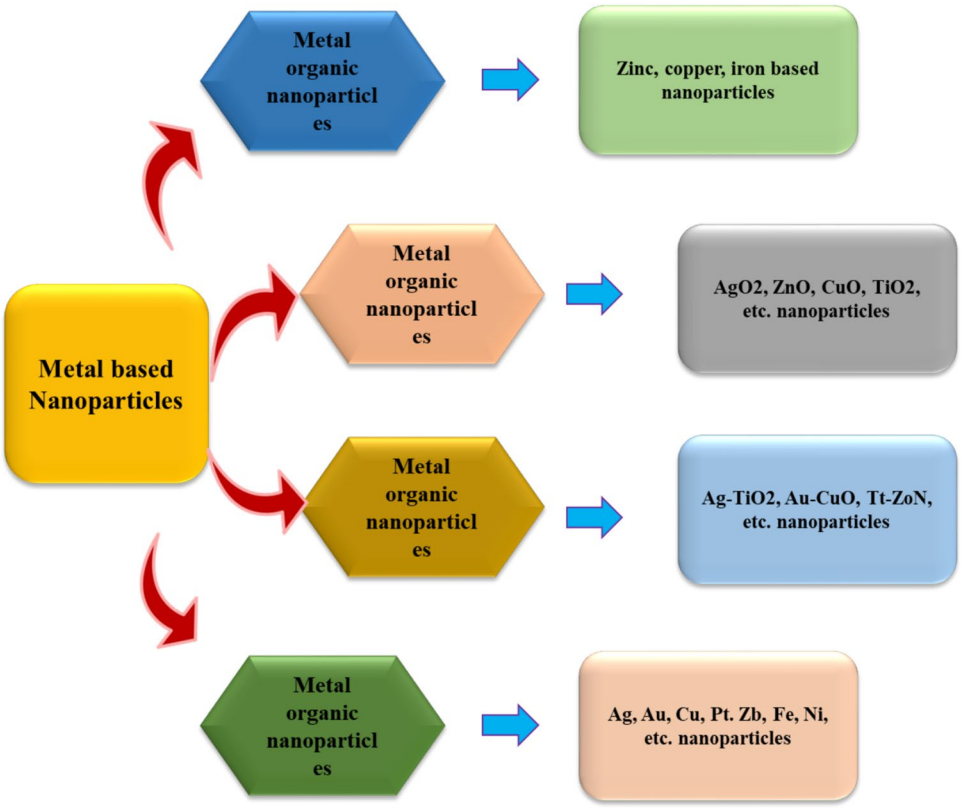

Fig. 3 Metal and metal oxide based on the nanoparticles

Table 3 Types of nanoparticles and their areas of applications

S. no	Nanoparticle type	Areas of applications	References
Carbon-based nanoparticles	Carbon nanotubes	Electronics, sensors, drug delivery, energy storage, aerospace	[20–24]
	Carbon nanofiber	Composites, electronics, energy storage, filtration, coatings	
	Fullerenes	Drug delivery, photovoltaics, water purification, lubricants	
	Graphene	Electronics, energy storage, composites, sensors, biomedical applications	
Metal-based nanoparticles	Silver	Antibacterial coatings, medical devices, wound care, electronics	
	Zinc	Antibacterial applications, sunscreens, food packaging, sensors	
	Iron	Magnetic resonance imaging (MRI), cancer treatment, wastewater treatment	
	Gold	Drug delivery, diagnostics, imaging, cancer therapy, biosensors	
Metal oxide-based nanoparticles	Zinc oxide	UV protection in sunscreens, electronics, photocatalysis, antibacterial coatings	
	Titanium oxide	Photocatalysis, water treatment, self-cleaning surfaces, sensors	
	Iron oxide	Magnetic storage devices, MRI, drug delivery, environmental cleanup	
	Cerium oxide	Antioxidant applications, catalysis	
	Silicon dioxide	Drug delivery, biosensors, electronics, photonic devices	
	Magnetite	Drug delivery, magnetic resonance imaging, environmental cleanup	

site of action, such as cancer cells, while minimizing damage to healthy tissues [113]. This targeted delivery system helps address the limitations of traditional drug delivery methods, such as poor solubility and nonspecific distribution. Moreover, certain nanoparticles can be engineered to release drugs in a controlled manner, triggered by environmental factors like pH, temperature, or external stimuli, ensuring sustained therapeutic effects and improving patient compliance [114].

Plant-Mediated Metal-Based Nanoparticles as Anticancer Agents

Metal-based nanoparticles, particularly those made from gold, have shown promise in anticancer therapy**.** Gold nanoparticles are effective in photo thermal therapy (PTT) and photodynamic therapy (PDT), where they can be targeted to cancer cells and activated by light to kill them. These nanoparticles are also used to deliver anticancer drugs and can enhance combination therapies and gene therapy, which targets specific genes that drive cancer cell proliferation [115]. Cancer remains one of the leading causes of death globally, and despite advances in genomics and proteomics, it continues to be difficult to treat due to its metastatic nature and drug resistance. Conventional chemotherapy often causes harm to healthy cells and has significant side effects, such as myelosuppression, microsites and organ dysfunction [116]. In contrast, plant derived biogenic nanoparticles offer a more targeted approach to cancer treatment of NPs showed in Fig. 4; Table 4. They can be functionalized with ligands or therapeutic molecules, improving the selectivity of treatment while minimizing side effects. These nanoparticles can also overcome barriers like the blood brain barrier, providing a potential treatment route for neurological cancers such as brain tumors [117].

Metal-Based Nanoparticles as Antimicrobial Agents

Metal-based nanoparticles, particularly silver**,** copper**,** zinc oxide and titanium dioxide, possess strong antimicrobial properties that make them effective against a wide range of bacteria, viruses, and fungi. These nanoparticles work by disrupting bacterial cell membranes, interfering with DNA replication, and generating reactive oxygen species (ROS), leading to microbial cell death [118, 119]. These nanoparticles are being integrated into wound dressings**,** medical device coatings and antimicrobial textiles to prevent infections and promote healing. Additionally, silver nanoparticles are commonly used in hospital settings for coating surfaces like bed rails and doorknobs to prevent the spread of infections [120].

Metal-Based Nanoparticles as Antioxidant Agents

Nanoparticles have shown significant potential as antioxidant agents, counteracting oxidative stress caused by free radicals. Free radicals contribute to various diseases, including cancer, cardiovascular diseases and neurodegenerative disorders. While the body naturally produces antioxidants, external sources are often necessary when antioxidant levels are insufficient [121]. Traditional synthetic antioxidants, such as BHA and BHT, can have harmful side effects, but metal-based nanoparticles can provide a safer and more effective alternative. The zinc oxide (ZnO) and selenium (Se) nanoparticles are emerging as biomedical antioxidants that protect cells from oxidative damage [121]. These

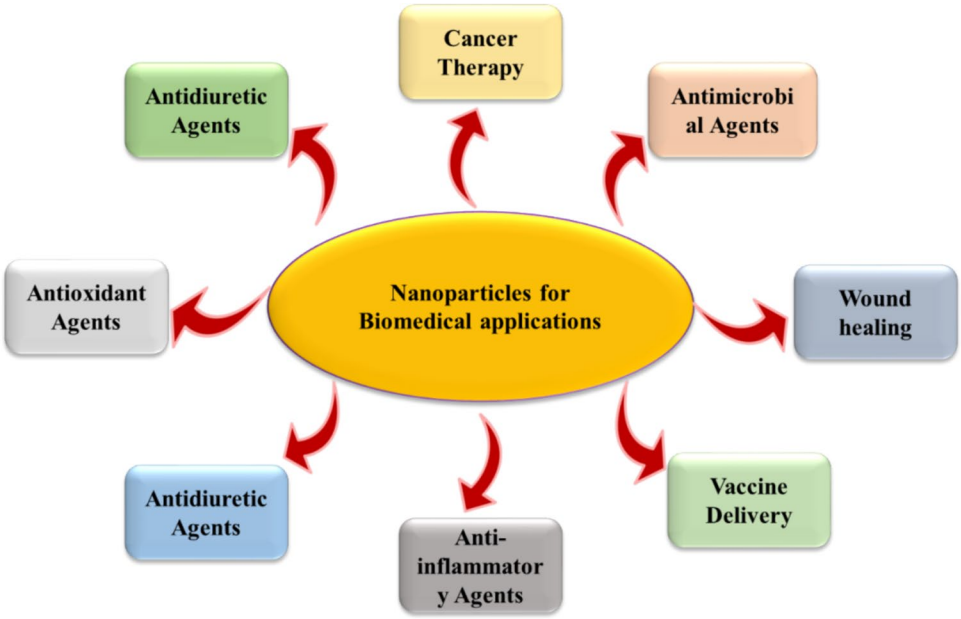

Fig. 4 Synthesized the nanoparticles biological applications

Table 4 Applications of organic nanoparticles

S. no	Organic nanoparticle type	Applications	References
1	Polymeric nanoparticles	Drug delivery, targeted therapy, gene delivery, cancer treatment, imaging	[20–23]
2	Lipid nanoparticles	Drug delivery (especially hydrophobic drugs), vaccines, gene delivery, cosmetic formulations	
3	Dendrimers	Drug delivery, gene therapy, diagnostic imaging, targeted therapy	
4	Micelles	Drug delivery, solubilization of poorly water-soluble drugs, targeted drug release	
5	Nanostructured lipid carriers (NLCs)	Drug delivery, controlled release, cosmetics, gene therapy	
6	Polymeric micelles	Drug delivery, tumor targeting, gene therapy, diagnostic imaging	
7	Solid lipid nanoparticles (SLNs)	Controlled drug release, topical drug delivery, cosmetic formulations	
8	Nanoemulsions	Drug delivery, food industry, cosmetic formulations, vaccine delivery	
9	Biodegradable nanoparticles	Drug delivery, tissue engineering, wound healing, cancer therapy	
10	Protein-based nanoparticles	Drug delivery, enzyme delivery, vaccine development, cancer treatment	

nanoparticles scavenge reactive oxygen species (ROS), thereby protecting DNA, lipids and proteins from oxidative damage, which can contribute to disease development. Functionalizing nanoparticles with antioxidants can enhance their ability to target specific oxidative stress markers, improving the effectiveness of treatments for diseases like inflammatory disorders and neurodegenerative diseases [122].

Metal-Based Nanoparticles as Anti-Inflammatory Agents

Chronic inflammation is associated with many diseases, including cancer, cardiovascular diseases and neurodegenerative disorders. Metal-based nanoparticles, such as zinc oxide (ZnO) and selenium nanoparticles (SeNPs), have been identified for their anti-inflammatory properties, which can help modulate inflammatory responses and treat conditions exacerbated by chronic inflammation [123]. In chronic diseases, traditional anti-inflammatory drugs (e.g., NSAIDs) are commonly used but can have side effects like gastrointestinal irritation and cardiovascular risks. Nanomedicine offers an alternative by improving drug delivery and targeting specific tissues, thus reducing the systemic side effects of conventional treatments. SeNPs, for instance, have been shown to reduce oxidative stress and inflammatory markers such as COX-2 and IL-6, making them suitable for treating conditions like arthritis, cardiovascular diseases and even neurological disorders [124].

Metal-Based Nanoparticles in Vaccine Delivery

Nanoparticles are being explored as vaccine delivery systems because they enhance stability, immune cell targeting

and the overall immune response. By acting as adjuvants, nanoparticles can increase the efficacy of vaccines by boosting the immune response and protecting the vaccine antigens from degradation [125]. This makes nanoparticles particularly useful for oral or nasal vaccine delivery, which is advantageous in areas with limited healthcare infrastructure. These nanoparticles can also be tailored for personalized medicine, delivering vaccines that target specific genetic mutations or strains, improving the overall effectiveness of treatments and reducing adverse reactions [126].

Metal-Based Nanoparticles in Agriculture

Nanoparticles hold considerable promise in agriculture, offering solutions for crop management, disease control and sustainability. Nanoparticles can improve nutrient bioavailability, soil health and pest control, making them valuable tools in addressing challenges such as climate change and food security [127]. Nano enhanced fertilizers encapsulate nutrients for controlled release, improving efficiency and reducing nutrient loss. This approach ensures that plants receive nutrients in a sustained manner, improving growth and reducing environmental pollution [71]. Similarly, nanoparticles can be used to boost crop yields, improve stress tolerance and enhance seed germination through better nutrient absorption [128]. Nanoparticles also offer innovative solutions for pest and disease control by delivering pesticides or insecticides directly to pests while minimizing the use of harmful chemicals [129]. Nano capsules and nanoparticle-based coatings can release pesticides only when ingested by pests, reducing environmental impact and improving the effectiveness of pest management strategies [74].

Nanoparticles in the Food Industry

In the food industry, nanoparticles are revolutionizing packaging, safety and nutrition. Nano cellulose and Nano clays are used as thickeners, stabilizers and texture enhancers, improving food quality and extending shelf life [130]. Additionally, nanoparticles are incorporated into smart packaging systems that can monitor food freshness, detect pathogens, and improve packaging material strength [131]. Nanoparticles also play a crucial role in fortifying foods with essential nutrients like vitamins, minerals and antioxidants, enhancing the bioavailability of these compounds and improving their absorption in the body [132]. Furthermore, biosensors are increasingly being used in the food industry to detect contaminants and pathogens, ensuring food safety and quality [133].

Nanoparticles in Environmental Remediation

Nanoparticles are integral to the field of bioremediation, where they are used to degrade or remove pollutants from the environment. Their high surface area and ability to interact with toxins make them effective in processes like wastewater treatment, soil restoration, and the removal of heavy metals from contaminated sites [134]. Metal based, non-metallic and carbon-based nanoparticles offer efficient solutions for environmental cleanup, leveraging their reactivity and high surface energy to break down pollutants or absorb contaminants [135].

Applications of Nanoparticles in Energy Technologies

Nanoparticles have gained significant attention for their role in enhancing energy technologies showed in Fig. 5; Table 5. Their unique size dependent properties, such as high surface area, quantum effects, and catalytic activity, make them ideal for boosting the efficiency, performance and cost-effectiveness of energy systems [136].

Quantum Dots (QDs) in Solar Cells

Quantum dots are nanoscale semiconductor materials with unique optical properties, including size-tunable light absorption and emission. This characteristic makes them highly effective in photovoltaic applications. In solar cells, quantum dots can expand the range of light absorption, allowing them to capture more of the solar spectrum and convert it into electricity [137]. By adjusting the size of the quantum dots, they can be engineered to absorb specific wavelengths of light, improving the efficiency of solar cells. Additionally, quantum dots can be used to enhance charge carrier movement, reducing energy losses and further boosting solar cell performance [138].

Plasmonic Photovoltaic with Silver and Gold Nanoparticles (AgNPs and AuNPs)

Silver (AgNPs) and gold (AuNPs) nanoparticles are widely used in plasmonic photovoltaic, where their surface plasmon resonance (SPR) properties enhance light trapping. When these nanoparticles interact with light, they create electromagnetic fields that can concentrate and redirect light toward the active layers of the solar cells, improving light absorption and reducing reflection. This enhances the overall efficiency of solar cells by allowing more light to be captured, leading to better energy conversion [139].

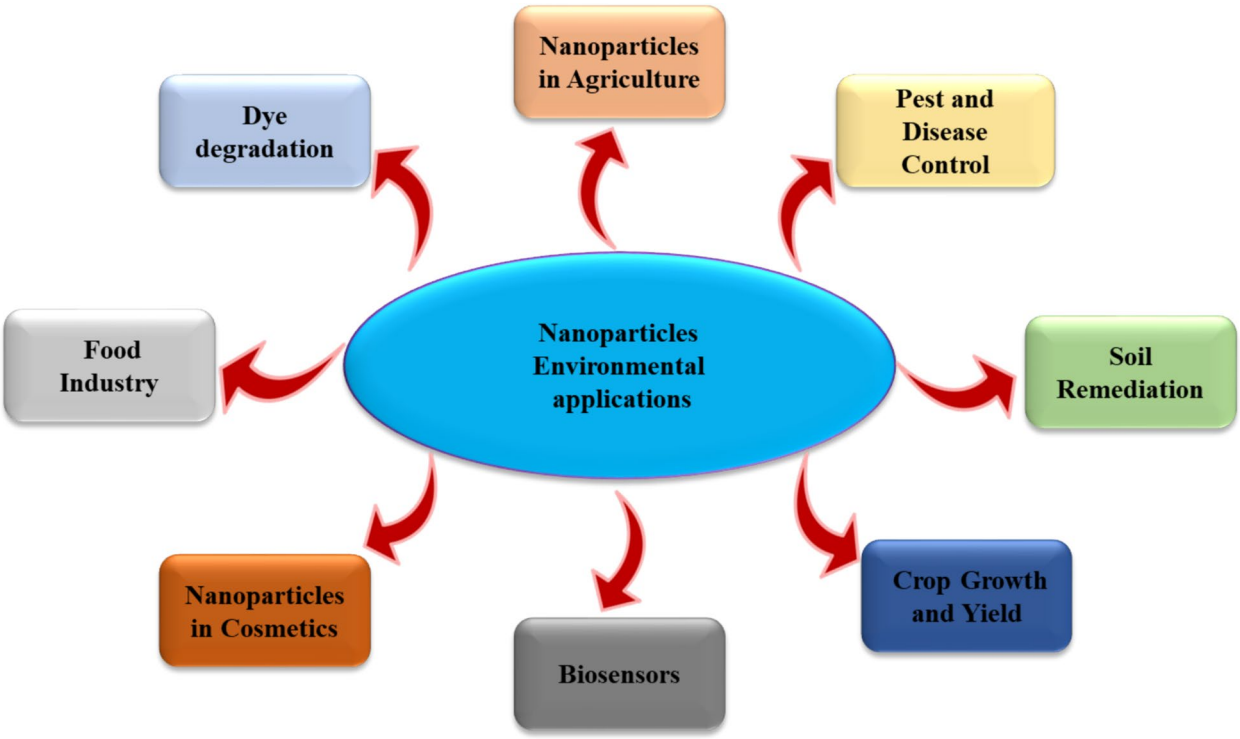

Fig. 5 Synthesized the nanoparticles in environmental applications

Table 5 Applications of inorganic nanoparticles

S. no	Inorganic nanoparticle type	Applications	References
1	Gold nanoparticles	Drug delivery, cancer therapy, diagnostic imaging, biosensors, photothermal therapy	[24–30]
2	Silver nanoparticles	Antibacterial coatings, wound healing, water purification, medical devices, biosensors	
3	Iron oxide nanoparticles (Magnetite)	Magnetic resonance imaging (MRI), drug delivery, cancer therapy, environmental cleanup, hyperthermia treatment	
4	Titanium dioxide nanoparticles	UV protection in sunscreens, photocatalysis, water treatment, self-cleaning surfaces	
5	Zinc oxide nanoparticles	Antibacterial applications, UV protection, sunscreen formulations, photocatalysis	
6	Silica nanoparticles	Drug delivery, biosensors, imaging, chromatography, cosmetics, food packaging	
7	Copper nanoparticles	Antimicrobial agents, drug delivery, environmental monitoring, catalysts	
8	Calcium carbonate nanoparticles	Drug delivery, food industry, tissue engineering, bone regeneration	
9	Silver-copper alloy nanoparticles	Antibacterial applications, wound care, medical devices, coatings	
10	Gold-silver alloy nanoparticles	Diagnostic imaging, cancer therapy, biosensors, photothermal therapy	
11	Platinum nanoparticles	Catalysis, cancer treatment, hydrogen storage, biosensors	
12	Palladium nanoparticles	Catalysis, hydrogen storage, sensor applications, environmental monitoring	

Titanium Dioxide (TiO_2) in Photo catalysis

Titanium dioxide (TiO_2) is one of the most commonly used photo catalysts for energy applications, especially in water splitting and carbon dioxide (CO_2) reduction. TiO_2 nanoparticles are capable of absorbing UV light and initiating reactions that split water molecules into hydrogen and oxygen, a process known as photo catalytic water splitting [140]. This hydrogen can be used as a clean fuel source. Additionally, TiO_2 is also used for CO_2 reduction, converting carbon dioxide into useful chemicals or fuels. The photo catalytic activity of TiO_2 can be improved by doping it with other elements or combining it with other nanomaterials, which enhances its light absorption and reaction efficiency [141].

Graphene and Derivatives in Photo catalysis

Graphene, a single layer of carbon atoms arranged in a hexagonal lattice, is known for its exceptional conductivity and high surface area. When combined with TiO_2 or other semiconductors, graphene and its derivatives can significantly enhance photo catalytic reactions [142]. The high surface area of graphene provides more active sites for photo catalytic reactions, while it's excellent electrical conductivity helps improve the transfer of charge carriers, reducing energy losses. This synergy enhances the efficiency of processes like CO_2 reduction and hydrogen generation via water splitting, making grapheme based composites ideal candidates for next-generation photo catalysis [143].

Silicon (Si) and Tin (Sn) Nanoparticles in Lithium-Ion Batteries

Lithium ion batteries (LIBs) are widely used in portable electronics and electric vehicles. However, one of the main challenges with LIBs is the degradation of the battery anode over repeated charge and discharge cycles, which limits their capacity and lifespan. Nanoparticles of silicon (Si) and tin (Sn) have been shown to significantly improve the performance of LIB anodes [144]. Silicon, for example, has a much higher capacity for lithium storage compared to conventional graphite anodes, but it suffers from large volume expansion during cycling. By using silicon nanoparticles, the effects of this expansion are minimized, improving the cycling stability and capacity of the battery. Similarly, tin nanoparticles improve the battery capacity retention and longevity. Sodium based nanoparticles are also being explored as more cost-effective alternatives to lithium, with the potential to further reduce the cost of battery production while maintaining performance [145].

Carbon-Based Nanoparticles in Super capacitors

Super capacitors are energy storage devices that offer rapid charge and discharge times, high power density and long cycle life. However, to improve their energy density, the conductivity of their electrodes must be enhanced. Carbon based nanoparticles, including carbon nanotubes (CNTs) and graphene, play a critical role in improving super capacitor performance [146]. These nanoparticles provide excellent electrical conductivity, a large surface area and high mechanical strength, which collectively boost the energy density and performance of super capacitors. In addition, metal oxides like manganese dioxide (MnO_2) and ruthenium oxide (RuO_2) are often incorporated into the electrodes to further enhance performance by increasing the capacitance and charge storage capacity [147].

Platinum (Pt) Nanoparticles in Fuel Cells

Fuel cells are devices that convert chemical energy from fuels like hydrogen into electricity through an electrochemical reaction. Platinum (Pt) nanoparticles are commonly used as catalysts in fuel cells due to their high efficiency in catalyzing reactions, such as the oxidation of hydrogen. However, platinum is expensive, limiting the widespread adoption of fuel cell technologies [148]. Researchers are exploring non-platinum alternatives to reduce the cost of fuel cells while maintaining or improving their performance. Nanoparticles of transition metals or alloys are being investigated for their ability to catalyze reactions effectively at a lower cost than platinum, making fuel cell technologies more accessible and cost-effective in the future [149].

Bismuth Telluride (Bi_2Te_3) and Silicon Nanoparticles in Thermoelectrics

Thermoelectric materials are used to convert heat into electricity, a process known as the thermoelectric effect. Bismuth telluride (Bi_2Te_3) nanoparticles are widely used for thermoelectric applications because of their high efficiency in converting heat to electrical energy. These materials are typically used in power generation from waste heat or for cooling applications [150]. Nano structuring Bi_2Te_3 can optimize its thermoelectric properties by enhancing the Seebeck coefficient, while reducing thermal conductivity, thus improving the efficiency of heat to electricity conversion. Silicon nanoparticles are also being explored for thermoelectric applications, as they offer a more abundant and cost-effective material compared to Bi_2Te_3. By optimizing the properties of these nanoparticles, more efficient thermoelectric devices can be developed for sustainable energy generation [151].

Environmental Impact of Nanoparticles

Nanoparticles (NPs) are increasingly being used in various fields such as agriculture, medicine, and industry due to their unique properties, including high surface area, reactivity and the ability to interact with biological systems. These properties have made nanoparticles highly valuable in a range of applications, from improving crop yields in agriculture to advancing drug delivery systems in medicine. However, as nanoparticles become more prevalent, understanding their behavior in the environment and assessing their potential ecological and health impacts has become a critical area of research [152].

Fate, Transport and Accumulation in Ecosystems

One of the main concerns regarding the environmental impact of nanoparticles is their persistence and movement through ecosystems. Because of their small size, nanoparticles are highly mobile in the environment, making it easier for them to travel through air, water and soil. This mobility can lead to their accumulation in various environmental compartments, including water bodies, soil and sediments [153]. In aquatic environments, nanoparticles can be absorbed by aquatic organisms, leading to bioaccumulation. Studies have shown that nanoparticles can enter the food chain through small organisms such as plankton, which are then consumed by larger organisms, including fish and humans. This bioaccumulation raises concerns about long-term ecological impacts and potential health risks. In soil, nanoparticles may affect plant growth and soil microorganisms, potentially altering nutrient cycling and soil health [153].

Toxicity to Nontarget Organisms

Another significant concern is the potential toxicity of nanoparticles to nontarget organisms. Due to their small size and high surface reactivity, nanoparticles can interact with biological systems in ways that larger particles cannot. When nanoparticles enter biological systems, they can cause oxidative stress, disrupt cellular functions and damage DNA. These interactions may lead to adverse effects in various organisms, including plants, animals, and microorganisms [154]. For instance, research has shown that certain nanoparticles can negatively affect aquatic life by disrupting the function of gills, intestines and other vital organs in fish and invertebrates. In plants, nanoparticles can be taken up by roots and affect cell structures, potentially inhibiting growth and affecting the plant ability to absorb nutrients. Additionally, nanoparticles can affect soil microorganisms, which are essential for nutrient cycling and soil fertility, potentially leading to long term ecosystem imbalances [155].

Exposure Pathways and Environmental Risks

The small size of nanoparticles enables them to cross biological barriers more easily than larger particles, increasing the risk of unintended exposure. Nanoparticles can be transported through water, air and soil and may enter ecosystems through industrial discharges, wastewater, runoff from agricultural land or atmospheric deposition. Once in the environment, nanoparticles can be ingested by organisms directly or indirectly, posing risks to both environmental and human health [156]. The nanoparticles used in agricultural applications, such as fertilizers or pesticides, can leach into water bodies, affecting aquatic organisms. Nanoparticles in cosmetics, cleaning agents, and other consumer products can end up in wastewater systems and eventually reach the environment. These exposure pathways underscore the importance of carefully managing and controlling the release of nanoparticles into the environment to minimize potential risks [157].

Need for Regulatory Frameworks

Given the increasing use of nanoparticles in various sectors, the development of comprehensive regulatory frameworks is essential to ensure their safe application and minimize environmental risks. Regulatory measures should focus on controlling nanoparticle release, assessing their environmental fate and ensuring that safety standards are in place for products that contain nanoparticles [158]. Currently, there is a lack of standardized protocols for assessing the environmental impact of nanoparticles and regulatory bodies are still in the process of developing guidelines. Comprehensive testing is needed to evaluate the toxicity, mobility, persistence and bioaccumulation potential of nanoparticles, with a focus on determining safe levels of exposure for both humans and wildlife. Regulatory frameworks should also consider the lifecycle of nanoparticles, from production to disposal, to ensure that their use does not lead to long-term environmental harm [116].

Optimizing Nanoparticle Formulations and Biological Interactions

Ongoing research is focused on optimizing nanoparticle formulations to mitigate potential risks. This includes modifying the surface properties of nanoparticles to make them less toxic and more easily degradable in the environment. For the surface chemistry of nanoparticles can be adjusted to reduce their ability to interact with biological systems or to enhance their biodegradability [159]. Additionally, researchers are investigating ways to control the size, shape and charge of nanoparticles, as these factors can influence their toxicity and environmental behavior. In the agricultural sector, efforts are being made to design nanoparticles that are biodegradable and non-toxic to nontarget organisms, ensuring that they provide benefits, such as enhanced nutrient delivery or pest control, without causing harm to the environment [118].

Nanoparticles in Precision Agriculture

Nanoparticles have shown great promise in precision agriculture, where they are used to improve the efficiency of nutrient delivery, pest control and soil health. Nanoparticles can be engineered to release nutrients in a controlled manner, ensuring that plants receive the required elements

without overuse or wastage [160]. This can help reduce the environmental impact of fertilizers, which are often over applied, leading to runoff into water systems and soil degradation. Moreover, nanoparticles can be integrated into smart agricultural systems, which use sensors and data analytics to optimize the use of resources. By targeting specific areas with the precise amount of fertilizers or pesticides, nanoparticles can help reduce the overall environmental footprint of agriculture. However, to ensure that these applications do not lead to environmental harm, it is crucial to monitor their effects and ensure that they do not accumulate in soil or water systems [161].

Ensuring Sustainable Nanotechnology Uses

The nanoparticles offer significant benefits across various fields, their potential environmental risks and safety concerns must be carefully managed. This includes ensuring the safe production, use and disposal of nanoparticles to minimize their release into the environment. In addition, further research is needed to understand the long term environmental and health impacts of nanoparticles, particularly as they continue to be integrated into new technologies and industries. The continued development of robust regulatory measures, combined with ongoing research into safer nanoparticle formulations and environmental monitoring, is essential to ensure that the use of nanoparticles remains sustainable and does not result in unintended harm to ecosystems or human health [162].

Current Challenges in Nanoparticles

Nanoparticles (NPs) have shown immense promise across various applications, ranging from medicine and agriculture to energy and environmental remediation. However, their widespread adoption faces significant challenges, primarily concerning safety, environmental impact, regulatory frameworks and production scalability. These challenges are further complicated by the unique properties of nanoparticles, such as their small size, high surface area, and reactivity, which can pose unforeseen risks to both human health and the environment [163].

Toxicity and Biocompatibility Concerns

One of the most pressing concerns regarding nanoparticles is their potential toxicity and biocompatibility. While nanoparticles, particularly those synthesized from plant based or green methods, offer several advantages, such as being eco-friendly and biocompatible, there remain significant risks associated with their use. Impurities from the synthesis process or alterations in the particles properties e.g., size, shape and surface charge can lead to unexpected toxic effects, especially in medical applications [123]. The nanoparticles can penetrate biological tissues more easily than larger particles, which raise concerns about their ability to accumulate in the body and cause long-term harm. These particles could potentially reach organs such as the liver, kidneys and brain, leading to health risks like inflammation, oxidative stress, or even cancer. Hence, rigorous and thorough toxicity studies are required to ensure that nanoparticles are safe for human use, particularly when incorporated into medical treatments such as drug delivery systems [164].

Persistence and Environmental Impact

Another major concern is the persistence of nanoparticles in the environment. Due to their small size and mobility, nanoparticles can easily disperse across ecosystems entering soil, water and air. Once released into the environment, they may persist for long periods, with uncertain long term consequences. Their ability to accumulate in organisms, including plants, aquatic life and soil microbes, raises the potential for bioaccumulation and bio magnification through the food chain [165]. The environmental risks posed by nanoparticles are particularly problematic when they are used in agricultural applications, such as in fertilizers or pesticides, where their runoff can contaminate waterways or disrupt local ecosystems. In addition, nanoparticles may adversely affect soil health by altering microbial communities essential for nutrient cycling. This environmental persistence could lead to disruptions in biodiversity and the overall health of ecosystems [106].

Lack of Standardized Protocols for Synthesis and Characterization

The synthesis, characterization, and application of nanoparticles still lack standardized protocols, which complicates research and development efforts. The diversity in nanoparticle types ranging from metals and metal oxides to carbon-based materials requires highly specialized techniques for their synthesis and analysis. However, the lack of universal standards for nanoparticle production leads to inconsistencies in particle size, shape, surface properties, and quality across different studies, making it difficult to compare results or scale up processes for industrial use [166]. Furthermore, the characterization of nanoparticles involves a range of sophisticated techniques, such as electron microscopy, dynamic light scattering and spectroscopy, but these methods are not always accessible or standardized across research labs. As a result, the reproducibility of nanoparticle based products and their functionalities is often uncertain and a lack of standardized protocols further hampers their integration into industries such as pharmaceuticals and agriculture [167].

Regulatory Gaps and Commercialization Barriers

The commercialization of nanoparticles is hindered by gaps in regulatory frameworks. Nanoparticles, particularly those used in consumer products, medicine and agriculture, require proper regulation to ensure that they are safe for public use. However, existing regulations often do not address the unique characteristics and potential risks posed by nanoparticles. For the nanoparticles might be classified differently from bulk materials and their behavior in biological systems is not always predictable based on traditional toxicity testing methods [168]. As a result, there is a need for specific regulatory standards that address the unique challenges of nanoparticles, including guidelines for their synthesis, environmental release and long term health effects. Without clear regulatory frameworks, nanoparticle based products may face delays in approval or face increased uncertainty in the marketplace. Moreover, concerns about the potential unknown risks associated with nanoparticles may deter investors and stakeholders from supporting the widespread adoption of nanotechnology [169]. In addition, the high cost of producing high quality nanoparticles particularly when scaling up production for commercial applications remains a significant barrier to widespread use. While plant based synthesis methods are cost-effective for small scale production, achieving consistent quality at larger scales remains challenging. Variations in size, shape and surface chemistry and stability issues can complicate the manufacturing process, leading to higher production costs and reduced reproducibility [170].

Public Concerns and Ethical Issues

Public concerns about the safety of nanoparticles, especially in consumer products such as cosmetics, food additives and cleaning agents, have raised significant ethical questions. One of the main concerns is the lack of long-term studies on the potential health and environmental effects of nanoparticles [171]. Ethical dilemmas arise when considering the uncertainty about their safety for human exposure and their potential to cause harm over extended periods. Issues related to privacy also arise, particularly in the use of nano enabled technologies for surveillance, tracking and data collection. The small size and potential for pervasive integration of nanoparticles into everyday items could lead to ethical concerns about individual privacy and consent. As nanoparticles become embedded in devices such as wearable health monitors, sensors, even smart textiles, privacy violations or unauthorized data collection could become a significant issue [172].

Opportunities in Green Synthesis and Nanotechnology

Despite these challenges, several advances in green synthesis of nanoparticles using natural resources, such as plants, fungi and bacteria, offer promising alternatives that mitigate environmental risks. Green synthesis methods often avoid the use of toxic chemicals, resulting in biodegradable, nontoxic nanoparticles that are more compatible with environmental and health standards. These eco-friendly synthesis approaches are expected to become more prominent as demand for sustainable products increases [173]. Green synthesis also opens new doors for the development of multifunctional nanoparticles. The agriculture, nanoparticles can be used to create controlled-release systems for fertilizers and pesticides, which minimize environmental damage and improve crop yield. Similarly, in medicine, advances in nanotheranostics and targeted drug delivery systems promise to provide more precise treatments with fewer side effects, revolutionizing the treatment of diseases like cancer [174].

Environmental Remediation and Sustainable Development

Nanoparticles have found applications in environmental remediation, including cleaning up oil spills and removing heavy metals from contaminated water. Research is increasingly focusing on developing nanoparticles that are not only effective in these processes but are also eco-friendly and sustainable. The nanoparticles could be designed to degrade pollutants into nontoxic compounds or adsorb harmful substances without leaving behind harmful residues [175]. As a nanotechnology progresses, researchers aim to create nanoparticles with multifunctional capabilities that could contribute to smart materials, such as self-healing surfaces or advanced biosensors, that will significantly impact future technologies. In this regard, nanoparticles could play a key role in sustainable development by providing solutions for clean energy, water purification and waste management [176].

Conclusion

In conclusion of this study, using medicinal plants for nanoparticle synthesis presents an eco-friendly, sustainable and cost-effective alternative to conventional chemical methods. Plant based synthesis utilizes phytochemicals that not only aid in reducing and stabilizing nanoparticles but also improve their efficacy in areas such as drug delivery and environmental remediation. As research advances, optimizing these synthesis processes and exploring new plant derived nanoparticle applications show significant promise. Green synthesis minimizes environmental risks, reduces costs and meets the increasing demand for sustainable

technologies. However, more research is needed to fully understand the mechanisms behind green synthesis and unlock its full potential. Current studies have produced nanoparticles with diverse structures, but there is a need for methods that create composite forms with enhanced surface areas and varied crystalline structures to discover new properties. Nanotechnology, operating at the nanometer scale, enables the development of materials with unique properties across numerous fields. Bio-green synthesis, using microorganisms or plant extracts, offers biocompatible and cost-effective nanoparticles for healthcare, particularly in cancer treatment, where traditional therapies often come with side effects. Further research into nanomaterials can also enhance food processing, bioremediation and wastewater treatment, playing a crucial role in addressing global challenges.

Acknowledgements The authors extend their appreciation to the Deanship of Scientific Research at Northern Border University, Arar, KSA for funding this research work through the project number "NBU-FFR-2025-1329-01".

Data Availability The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declared that there is no conflict of interest.

Research Involing Human and Animal Rights All authors hereby declare that there are no ethical concerns related to human or animal rights in this study.

Consent for Publication All the authors agreed to publish the data in this journal.

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